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Process modeling for robotic polishing

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Abstract

This paper presents the solutions adopted for a robotic polishing cell for mold manufacturing. Mold finishing is frequently carried out manually; these kinds of operations are iterative, time consuming and require experience. Automation can introduce cost reduction minimizing production times on such manual finishing operations. The research shown in this paper provides an overall solution without discarding aspects that could be relevant to the task considered in the mold making environment. Most of the mold makers usually employ CAD/CAM systems in order to design workpieces and to prepare the CNC programs to manufacture the parts, this CAD data can also be used to prepare programs for the actual robot finishing of these parts. In this work, an automatic planning and programming system based on data from a CAD system is described. The direct use of the paths implies difficulties of adaptation mostly when free-shaped surfaces are treated, so that the use of a constant contact force control has also been considered. To accomplish the automation of the process it is necessary to establish a behavior model of the system's variables. This fact allows the definition of the task specifications of the tools and, as a result, both the planning system and the control strategy by stages of the polishing process.
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Keywords: Polishing; Robot; Automation

1. Introduction

The objective of this paper is to present a description of the practical solutions adopted for building a robotic polishing cell for molds and free-form surfaces, the modeling of the polishing process, and its advantages in comparison with traditional methods used for this kind of part. Automation of the polishing process requires four basic development steps: (1) tool paths generation is obtained from a CAD system; (2) a controller with force feedback is needed to maintain constant tracking on real surface; (3) the polishing process is modeled by means of a computer application in order to predict the roughness values reached during the finishing of the surface; (4) this application also provides algorithms for controlling the process in order to reach the target finishing grade.

Nowadays, most of the finishing processes over complex surfaces are carried out manually. The execution of these processes is highly based on the expertise of the operator who carries them out, and moreover, this kind of processes are frequently poorly modeled. The automation of the finishing operations represents the major technological barrier to reduce production costs.

Several approaches from different research groups have been trying to automate these processes. From the research works carried out in this field, it is meaningful to point out those made by Tönshoff et al. [1] who described a state-of-the-art technique in modeling and simulation of grinding processes. A theoretical model generalization on different aspects of the grinding process was proposed. Kurfess and Whitney [2] presented a predictive controller to interface with a robotic weld bead grinding system. To carry out this, a grinding model was used, this model was based on a nonlinear first-order differential equation. Kasai et al. [3] completed experimental work on polishing of optical devices and metallic surfaces, employing two roughness formation models to justify the use of flexible abrasive supports instead of rigid ones. Saito et al. [4] carried out the development of an expert planning system of the polishing process in order to establish an automatic manufacturing system for dies and molds. This system tries to be useful for schedule the polishing process and to be as similar as possible to expert performance of mold machinist. Mizugaki and Sakamoto [5] suggested a method of tool path generation from CAD system for a robot polishing system. The system was based on flat fractal curves generation and projection onto the free-form surface of a workpiece, to obtain a theoretical tool path over the surface.

In other works about systems of automated polishing, Mizugaki et al. [6] presented one of the earliest robot systems with contact pressure control using CAD/CAM data.

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Nomenclature

c	exponent of the finishing grade evolution curve
e_1, e_2, e_3	components of the vector $\vec{e}$
$\vec{e}$	unit vector to define an orientation of the tool
F_x, F_y, F_z	components of the polishing force in X-, Y-, Z-axes
HB	workpiece hardness (Brinell scale)
i	counter for the number of operations
$\vec{i}$, $\vec{j}$, $\vec{k}$	components of the Cartesian unit vector in X-, Y-, Z-axes
j	counter for the number of passes
$\vec{n}$	unit vector normal to the surface
n_x, n_y, n_z	components of the vector $\vec{n}$
N_{it}	number of passes to reach R_{ait}
q	quaternion
q_i	quaternion components
R_a	general finishing grade measured by the arithmetic mean deviation roughness
R_{ai}	finishing grades reached in polishing operation number i
R_{ait}	practical value where R_{mi} can be considered is reached
R_f	final finishing grade after the complete set of operations (target finishing grade)
R_{fi}	final finishing grade reached after the operation number i
R_{mi}	minimum reachable finishing grade for the operation number i
R_o	initial finishing grade at the beginning of the polishing process
R_{oi}	initial finishing grade at the beginning of the operation number i
S_x, S_y, S_z	components of the dynamometer sensitivity in X-, Y-, Z-axes
TG_i	available abrasive grain size for the operation number i
TG_{if}	minimal available abrasive grain size
Greek letters	
ΔN_m	minimal number of passes between roughness measurements
θ	angle to define the rotation about the unit vector

Kuo [7] used fuzzy neural networks for a robotic die polishing system. Ahn et al. [8] also used a expert system for optimizing the process. Lee et al. [9] used CAM data to control the trajectories of a polishing robot system. Su and Sheen [10] developed a process planning strategy for a very particular application. And finally, Tam et al. [11] accomplished the problem through scanning paths to polish free-form surfaces.

About the polishing processes and its variables, there is a great number of works, of which some of the latter ones are those of Lin and Wu [12] that studied the effects of polishing parameters on material removal rates and non-uniformity effects, Klamecki [13] built up a comparison of material removal rate models and experimental results in double sided polishing processes, Kim et al. [14] worked with active profiling for the polishing of large precision surfaces with moderate asphericity, Su et al. [15] made a study on smoothing efficiency of surface irregularities by hydrodynamic polishing, Hocheng and Kuo [16] and Zhao et al. [17] used the ultrasonic polishing for mold steels and free-form surfaces, and Venkatesh et al. [18] made a theory about the genesis of workpiece roughness generated in surface grinding and polishing of metals.

These works provide different partial views of the aspects involved in the complex surface polishing problem. The proposed automatic robotic polishing system intends to integrate one theoretical model for finishing parameters evolution with a control system for a robotic polishing cell to automate as far as possible the finishing operations over complex geometry workpieces. To carry out this objective one process planner and one computer aided programming system have been developed within of a CAD system environment. These applications have been presented in previous works of Marquez et al. [19,20].

2. Experimental setup

The cell is based on a six axis robot, located in front of the workbench in which the molds to polish are fixed. On the left-hand side of the robot the motorized tool storage is located. Fig. 1 depicts a layout of the polishing cell. The cell is made up of an ASEA ABB IRB-3000 robot, with the control system S3 and a personal computer where the supervision system as well as the stage planner and control system are executed. The CAD system (Autocad v14 or Euclid v3) is running on a separate workstation.

The force control instrumentation is made up of a three axis piezoelectric force sensor Kistler 9257A with force

Fig. 1. Robot polishing cell experimental setup.

ranges of $F_x = 0$–5000, $F_y = 0$–5000, $F_z = 0$–10 000 N, and sensitivities of $S_x = 7.9$, $S_y = 7.9$, $S_z = 3.8$ pC/N. The finishing evaluation is done by one roughnessmeter Hommel T-500 and one reflectometer Dr. Lange Refo-3M.

3. Finishing evolution model

The evolution of the polishing process of a workpiece is considered as a succession of finishing states. In this work, the finishing state is evaluated by the R_a roughness parameter (arithmetical mean deviation). The measurement of two quantities, roughness and brightness, provides the state information needed to execute the process control. A combined system to evaluate the finishing state has been selected in order to simplify future applications of the system. Roughness measurement is difficult in the presence of vibrations or curved regions where an optical system could provide better results. Although the instrumentation employed is difficult to implement in a completely automated system, a considerable effort has been undertaken in order to develop an open architecture to permit the integration of future improved instrumentation techniques.

3.1. Process hierarchy

As result of their manual execution, polishing processes are not described in a structured form. For automation it is

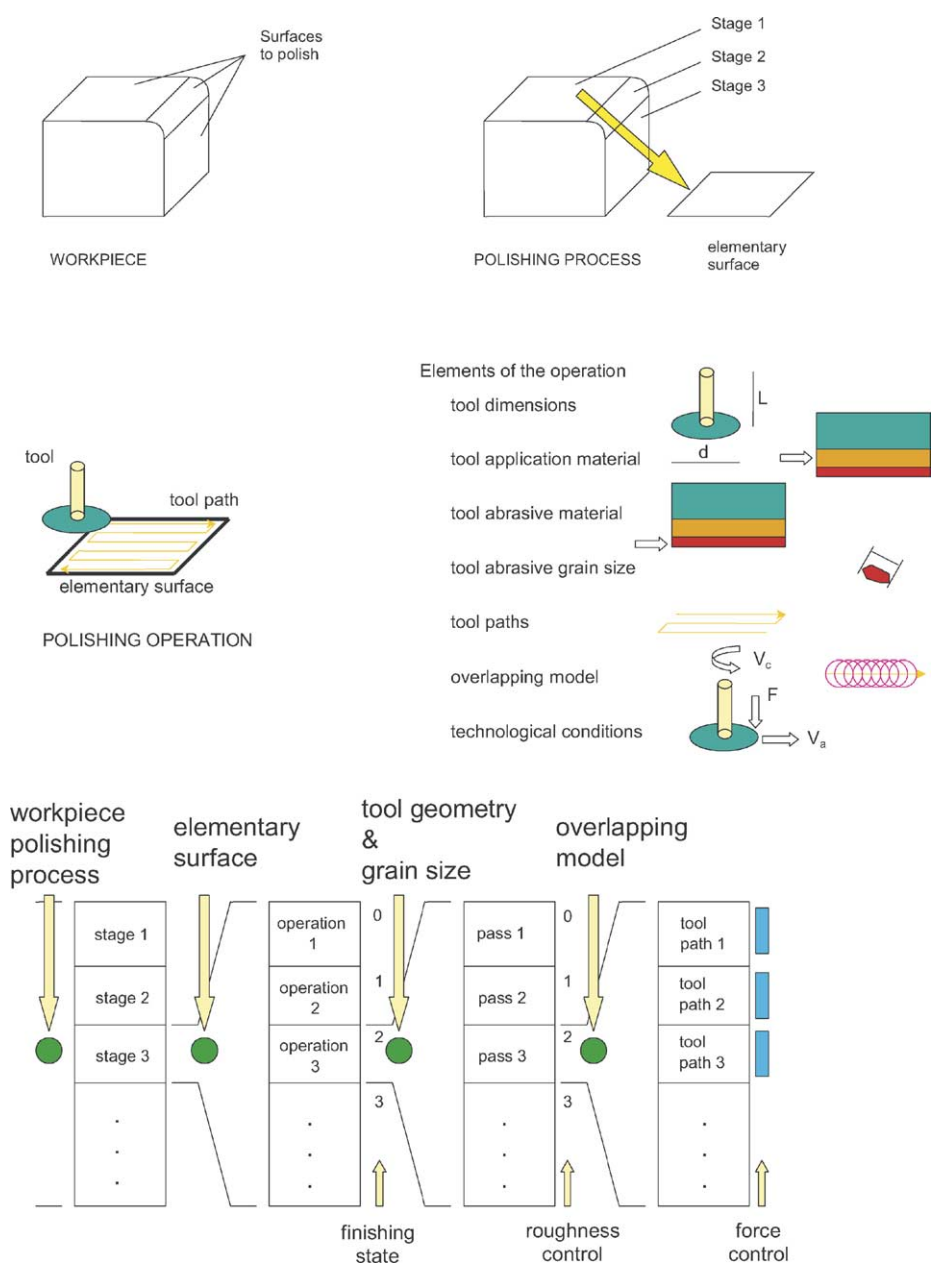


Fig. 2. Elements of the polishing model.

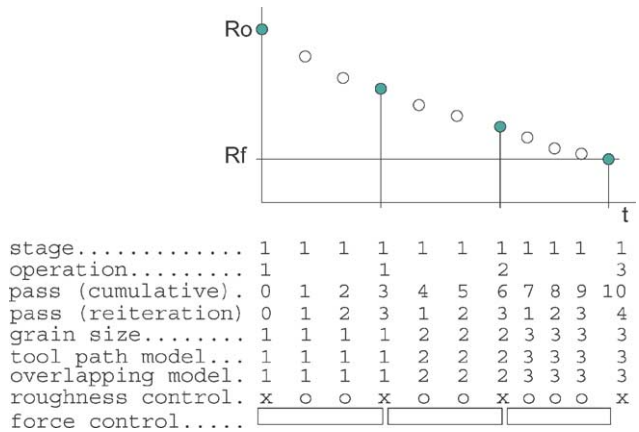


Fig. 3. Finishing evolution.

necessary to define a formal multistage hierarchy. In this section a brief definition of a formal process hierarchy is provided (Figs. 2 and 3).

3.1.1. Process

Polishing process is a set of polishing stages to transition from an initial finishing state of a workpiece to the desired final finishing state of the workpiece surfaces to be machined. It is supported on the CAD system.

3.1.2. Stage

The polishing stage is a set of polishing operations to transition (over one single surface of the workpiece) from an initial finishing state to the desired final finishing state. The term “surface” represents the main difference between the first definition that is related to the entire workpiece. As a consequence one polishing process is a set of polishing stages over various surfaces of the workpiece. It is performed on the PC that controls the robot.

3.1.3. Operation

The polishing stage is not a continuous phase, in fact it is composed of different “polishing operations” attending to particular geometric or functional criteria such as tool geometry or abrasive grain size. One polishing operation can be defined as a set of polishing passes (with uniformed geometric and functional specifications) to transition from one initial or intermediate finishing state to another intermediate or final finishing state. The main parameters under control are the finishing degree and the tool force. It is performed on the PC that controls the robot.

3.1.4. Pass

The polishing pass is the minimal process unit and is defined as a set of tool paths described with continuous contact over the surface within one polishing operation. Thus, one polishing operation can be composed of multiple polishing passes. The main parameter under control is the contact force between tool and workpiece. It is performed on the PC that controls the robot.

3.2. Application levels for the model

The software applications that develop the model have been divided in areas that are described in the following: process level and stage level, and both of them are named as the stage planner.

3.2.1. Process level

At the process level is necessary to establish the different stages in which the polishing process of a workpiece is going to be subdivided. Therefore, a stage planner and an automatic toolpath programming system is developed for this model application. Both of them have been developed in a CAD environment in order to facilitate the management of the workpiece geometry.

Most of industrial robots are learning programmed, this fact implies difficulties when complex geometry as free-form surfaces are treated. For mass production this really does not represent a major problem as it does in short or prototype series where it is critical to minimize programming time. For this research, a graphical CAD-based programming approach has been chosen. The data provided by the CAD system is used in the computer application developed to carry out the polishing stage planning and the operations programming. Fig. 4 depicts the schematic plan for the mold process elaboration.

The stage planner is based in the association of a specific surface in the workpiece with a tool and a set of technological parameters needed to define operations and passes of the polishing process. Each stage is going to be related with a specific workpiece surface and with a specific tool geometry to develop the successive operations. After the selection of the tools, the tool path is generated. Different approximations give as a common result the need of a great amount of overlapping in trajectories. Fig. 5 depicts the solution adopted for this research based on experimental results.

As a consequence of the nature of the polishing process, where the finishing states are obtained by means of multiple material removing operations, accomplished in different directions so that there is not a predominant direction in which irregularities are greater than in the other ones; it is necessary to define tool paths trajectories in such a way that the crossing of tool displacements reaches a minimum value. This is known as overlapping.

Fig. 5 depicts different simple alternatives used in this research. When selecting the overlapping model there are two possible alternatives: use a simple overlapping model with a high number of iterations; or to use a more complex with few iterations. There is not a significant difference in the experimental results. The third is a more straightforward model especially over complex surfaces, and shows better control along the evolution curve of the finishing grade versus time or number of passes. In Fig. 5 tool trajectories are shown that are composed of two elements: the tool path that defines the exploration lines over the part surface as a grid; the overlapping model created from basic units that are

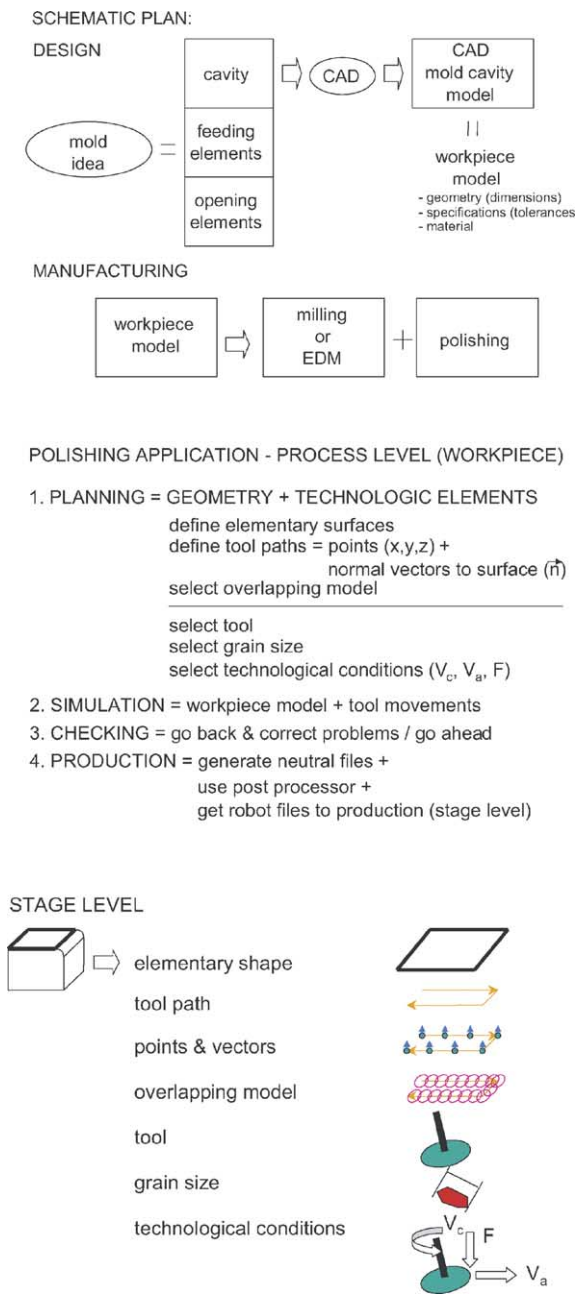


Fig. 4. Schematic plan for mold process.

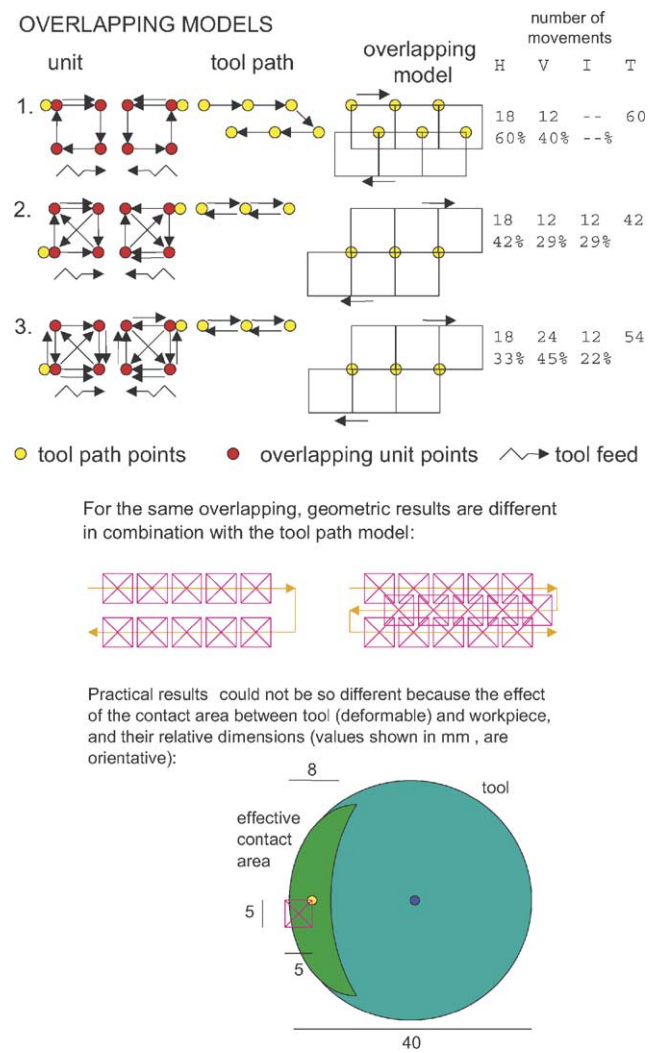
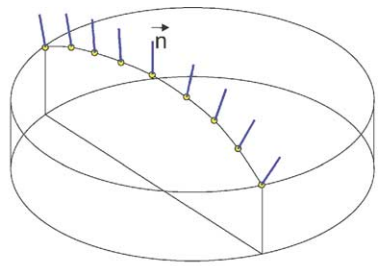


Fig. 5. Overlapping models.

repeated along the former tool paths. Both of them are defined by points: tool path points; overlapping model unit points. As a comparison among different overlapping models the following parameters are computed in absolute and percentage terms: number of movement segments (H: horizontal; V: vertical; I: inclined; T: total) for each overlapping model (and for six units in this figure).

Neutral files are used as a common output in computer aided NC and robot programming, where geometrical and technological information is stored, to feed any postprocessor unit that generates the final code. These files hold the information about tool path geometry, overlapping model and technological parameters such as feed rate and cutting speed, and can be generated from CAD application data. Fig. 6 shows an example of the tool path generation over a spherical surface with indications of the vectors normal to the surface in each of the points of interpolation. Positions are given to the robotic system with the appropriate orientation in quaternions.

The number of arithmetical operations to give a six axis robot the required three-dimensional orientation can be drastically reduced employing Euler parameters (quaternions) instead of Euler angles. This simplifies considerably the algorithm employed for the transformation of normal vectors to the surface in robot arm orientations. The representation in Euler parameters is based on Euler theorem which establishes that the displacement of a rigid body with a fixed point can be described as a rotation around one common axis to both reference systems. The Euler parameters are a representation in four component arrays in which three components represent the rotation axis and the fourth one the angle rotated. The Euler parameters calculation in our particular case is explained in the following.



n = normal vector to surface in each tool path

∘ = interpolation points for define the tool path calculated by intersection of parallel planes normal to the base plane, and the surface to be polished

point	x	y	z	q0	q1	q2	q3
1	1427.57	-48.16	796.13	.623827	.0	.734192	.204433
2	1427.57	-18.68	801.44	.623827	.0	.759335	.064792
3	1427.57	11.28	801.20	.623827	.0	.758186	-.077101
4	1417.57	44.94	796.13	.647684	.0	.715729	-.230607
5	1417.57	15.70	802.66	.647684	.0	.745863	-.095597
6	1417.57	-14.34	803.71	.647684	.0	.750747	.043124
7	1417.57	-44.38	799.14	.647684	.0	.729634	.181860
8	1407.57	-59.09	796.13	.670649	.0	.696816	.243161
9	1407.57	-30.02	803.38	.670649	.0	.729417	-.021949

Fig. 6. Tool path automatic generation.

The Euler parameters q have a four component representation that can be written as an scalar component q_0 and one vector $\vec{q}$:

$$q = q_0 + \vec{q} = q_0 + q_1\vec{i} + q_2\vec{j} + q_3\vec{k} \tag{1}$$

$$q = (q_0, \vec{q}) \tag{2}$$

where $\vec{i}$, $\vec{j}$ and $\vec{k}$ are the unit vectors in X'-, Y'- and Z'-axes, respectively, where these axes are the reference system X, Y, Z translated to the tool.

The Euler parameters to determine the relative rotation θ about one common axis whose direction is defined by a unit vector $\vec{e}$ was defined by Euler:

$$q = \cos\frac{\theta}{2} + \vec{e}\sin\frac{\theta}{2} \tag{3}$$

using the $\vec{e}$ components:

$$q = \cos\frac{\theta}{2} + (e_1\vec{i} + e_2\vec{j} + e_3\vec{k})\sin\frac{\theta}{2} \tag{4}$$

where e_1, e_2 and e_3 are the director cosines of $\vec{e}$. The Euler parameters can also be written as

$$q = q_0 + \vec{q} = \cos\frac{\theta}{2} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}\sin\frac{\theta}{2} \tag{5}$$

The main problem is to pass from determined directions in the robot wrist reference trihedral to those actual directions of trihedral in order to put the robot normal to the desired workpiece surface. The Euler theorem establishes the existence of one common axis to both trihedral and the existence of one direction vector which magnitude contains information about the rotated angle between trihedrals X', Y', Z' and X'', Y'', Z''. The objective is to evaluate the new orientation Euler parameters as a function of the preexistent surface normal vector components.

$$\vec{n} = n_x, n_y, n_z \tag{6}$$

where n_x, n_y, n_z are the absolute reference trihedral components located in the robot.

The common vector to both reference relative trihedrals (successive orientations in a program) can be easily calculated aligning the positive direction of axis X'' opposite to $\vec{n}$ (Fig. 7):

$$\vec{e} = -\vec{n} \times \vec{i} \tag{7}$$

The expression for $\vec{e}$ can be reduced to

$$\vec{e} = \begin{bmatrix} \vec{i} & \vec{j} & \vec{k} \\ -n_x & -n_y & -n_z \\ 1 & 0 & 0 \end{bmatrix} = -n_z\vec{j} + n_y\vec{k} \tag{8}$$

The rotation angle can be expressed as

$$\theta = \arccos\left(\frac{\vec{n}\cdot\vec{i}}{n\cdot i}\right) = \arccos(-n_x) \tag{9}$$

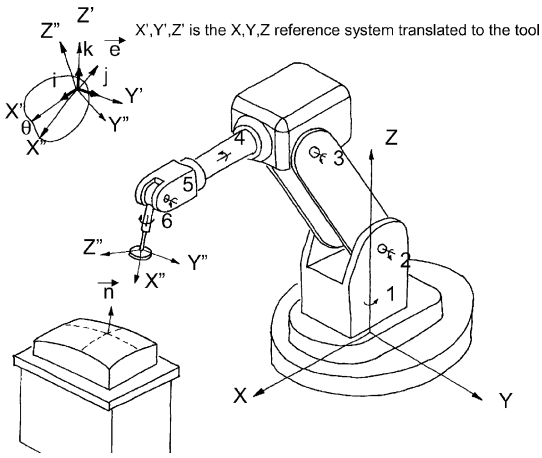


Fig. 7. Coordinate systems and reference vectors.

With these calculated values, the expression for the Euler parameters can be evaluated:

$$q = q_0 + \vec{q} = q_0 + q_1\vec{i} + q_2\vec{j} + q_3\vec{k} = \cos\frac{\theta}{2} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} \sin\frac{\theta}{2} \tag{10}$$

The expressions for the Euler parameters can be particularized as

$$q_0 = \cos\left(\frac{\arccos(-n_x)}{2}\right) \tag{11}$$

$$q_1 = 0 \tag{12}$$

$$q_2 = -n_z \sin\left(\frac{\arccos(-n_x)}{2}\right) \tag{13}$$

$$q_3 = n_y \sin\left(\frac{\arccos(-n_x)}{2}\right) \tag{14}$$

3.2.2. Stage level

At this level the appropriate use of an evolution model for the roughness is needed, in order to achieve the correct finishing state for each stage. The model has been inferred from experimental data obtained in polishing tests over different workpieces carried out with many different tools.

The recursive exploration of the elementary surface with the abrasive tool, gets the progressive reduction of the finishing grade according with a decreasing exponential curve along the machining time or the number of passes.

The parameters of the exponential curve are obtained from a set of experimental setups. These values are used in the polishing stage control. For optimal results, the conditions used in the experiments must be similar to those applied in the stage. This evolution curve has been used by in previous researches such as Saito et al. [4].

The evolution curve for the roughness reached (R_{ai}) along the number of passes (j), with continuous machining maintaining the same abrasive grain and tool, is based on a exponential equation, Fig. 8, used in an operation where the grain size and tool geometry are maintained constants.

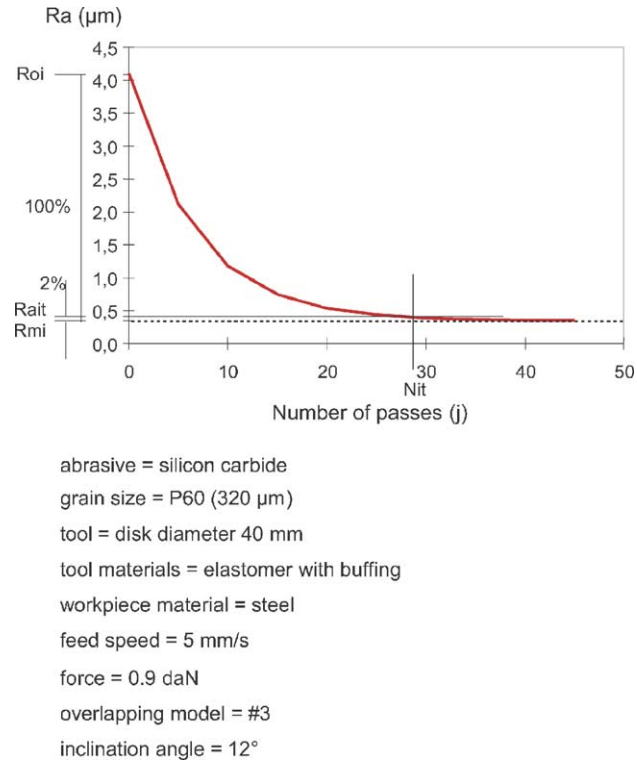


Fig. 8. Exponential model for a polishing operation.

$$R_{ai} = (R_{oi} - R_{mi})\, e^{-cj} + R_{mi} \tag{15}$$

where, as it is usual in machining processes, the finishing grades are associated with the arithmetical mean value of roughness. For clarity, every roughness is denoted beginning with the letter R, instead of R_a, except the general term that varies along the process. Some significant global parameters are: R_o the initial finishing grade or roughness (that it is obtained in the previous machining process, milling typically); R_f, the final roughness to be obtained by the polishing process as it is defined in the design specification; R_m, the better roughness obtained with an specific grain size.

Each of the successive polishing operations (denoted by the index i) are defined by R_{oi} as the initial roughness (that has the same value as R_{fi-1}), R_{fi} the roughness reached in this operation (that has the same value as R_{oi+1}), and R_{mi} as the minimum reachable roughness value (asymptotic) for the grain size used. Since this value it is difficult to obtain in periods not too long, it is defined a practical limit R_{ait} where it can be considered that R_{mi} has been reached in practical terms. This value is defined by means of an allowance, typically considered as 2%, and so, the final roughness values in each operation R_{fi} are limited as $R_{mi} \le R_{fi} \le R_{ait}$.

This equation must be applied for each abrasive grain size employed; thus, the stage will have different exponential evolutions associated with the different grain sizes employed

Table 1
Minimum reachable roughness and exponent

Abrasive code	Mean grain size (μm)	R_m (μm)	c (HB170)	c (HB110)
60	425–355	0.35	0.150	0.135
100	180–150	0.33	0.161	0.145
120	150–125	0.30	0.172	0.155
180	90–75	0.28	0.183	0.165
240	63–53	0.25	0.194	0.175
320	48–44	0.15	0.205	0.185
400	37–33	0.10	0.216	0.195
600	27–24	0.08	0.227	0.205
5	5	0.05	0.238	0.215
1	1	0.02	0.250	0.225

for each operation. In order to obtain different evolution curves, an estimation of exponent values c, and asymptotic values R_{mi}, are obtained for each abrasive grain size.

Table 1 depicts minimum reachable roughness R_{mi} for different grain sizes, and reflects the evolution of the exponent, c, as a result from the test performed for different abrasive grain sizes with rubber tools with buffing over steel workpieces. The experimental estimation of these parameters must be as reproducible as possible to enable application to other automated systems.

The procedure to define the sequence of operations for the workpiece in a polishing stage begins with the initial data input for the stage planner (Fig. 3). The operation sequence is defined in terms of a sequence of successive abrasive grain sizes.

Table 2 depicts four operational sequences predicted by the stage planner with different input data. The stage is composed of a variable number of operations to reach the final roughness. For example, stage 1 is indicated on a workpiece with HB170 base material hardness and an initial R_o roughness of 4 μm, and it is composed of six polishing operations from initial abrasive grain size code 60 (425–355 μm) silicon carbide, to 1 μm aluminum oxide. Each operation change implies an abrasive grain size change to reduce the minimum reachable roughness value associated with this grain size.

Table 2
Stage planner output

	Stages			
	$R_o > 2$ μm		$R_o < 2$ μm	
	1	2	3	4
Hardness (HB)	170	110	170	110
Abrasive code				
R_m (μm)				
0.35	60			
0.33		100		
0.30			120	
0.28				180
0.25	240			
0.15		320		
0.10	400	400	400	400
0.08	600	600	600	600
0.05	5		5	
0.02	1	1	1	1

The theoretical evolution of the stage can be evaluated according to the developed exponential model. The stage planner selects the minimum reachable roughness value and c. With this data it represents the theoretical curves for the whole stage. The exponential curves for stage 1 predicted by the stage planner can be readily observed (Figs. 9 and 10).

4. Control strategy

The modeling of the polishing process has revealed two elements that directly affect the implementation of the control system on the robot cell: operation control where the finishing values are checked, and the polishing pass control where the operational conditions are applied.

4.1. Operation control

There is a physical drawback in getting the minimal reachable roughness related to the abrasive grain size used. This fact requires an estimation of the number of operations that are needed to reach the target roughness R_f. The goal of the stage planner is to adjust the task specifications; however, the predictions of the polishing process are poor if we compare them with other classic machining processes. Therefore, the system must prevent model deviation and support the proper control structure. Fig. 11 depicts the control hierarchy used in the proposed system.

The objective of the operation control is to allow the stage evolution in case of unpredictable behavior in the model. The operation control is based on the model for the roughness evolution. Fig. 12 presents a graph relating roughness to number of passes depicting various possibilities that can occur during a polishing operation.

The different possibilities can be reduced to four basic types of evolution models:

- *Type A*: Behaviors that reach the desired roughness before predicted by the model. The intersection with R_{mi} (minimal roughness reachable with this abrasive grain size) is achieved before predicted time and the system should detect it and continue with the next operation within the stage.
- *Type B*: Behaviors in which the system has not reached the roughness limit after the predicted number of passes. The strategy in this case is to continue conducting polishing passes until this value is reached.
- *Type C*: The operation control is going to allow a number of passes greater than the theoretical prediction. However, this number of passes must be finite. As it is possible that the minimal roughness reachable with this abrasive grain size cannot be achieved due to cost or time constraints, a limit on the number of passes is set. The possible behavior shown for C predicts that maximum number of passes

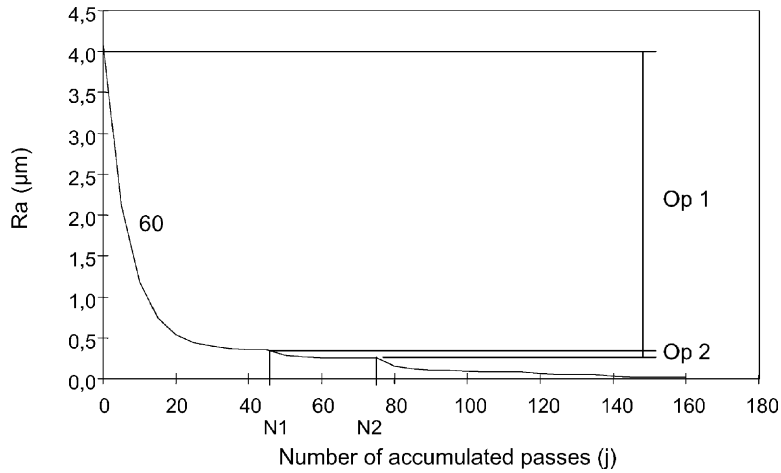


Fig. 9. Theoretical polishing operations (operations 1 and 2).

should be not greater than *n* times the theoretical number of passes (this number *n* is fixed according the grain size), above this number the theoretical roughness is considered as unreachable using this abrasive grain size and so, grain size is reduced.

- *Type D*: This case is when there is a significant deviation from the model predictions for the minimum roughness reachable with this abrasive grain size with respect to the measured roughness. The strategy in this case is to change the abrasive grain size, as it is not possible to reach the operation objective with the current grain size.

The control scheme will be based on the periodic measurement of the finishing state to make decisions according to the strategies defined for the different operation evolution types.

A flow diagram is presented divided for clarity in Figs. 13 and 14 to show the inference algorithm for the operation control system. Fig. 13 shows algorithm for control evolution models A and E, and Fig. 14 for B–D.

The experimental results match directly with the predicted behavior, as it is shown in Fig. 15, and sometimes present irregularities such as those shown in Fig. 16, where due to inaccuracies in the definition of the technological conditions, finishing gets worse in a number of polishing passes. Thanks to the recorded data, this is a valuable method to improve the efficiency of the application for further polishing processes through improved technological conditions data.

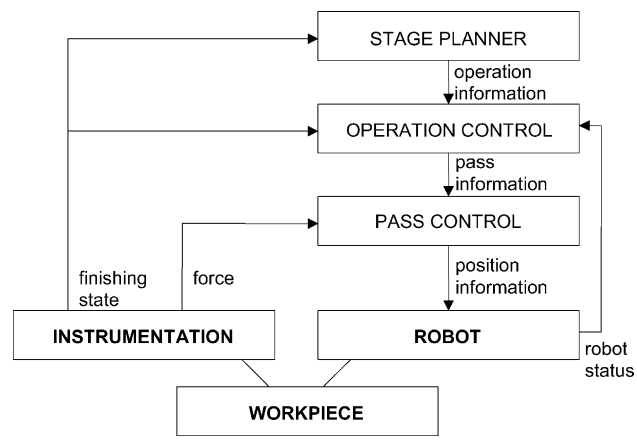


Fig. 11. Process control hierarchy.

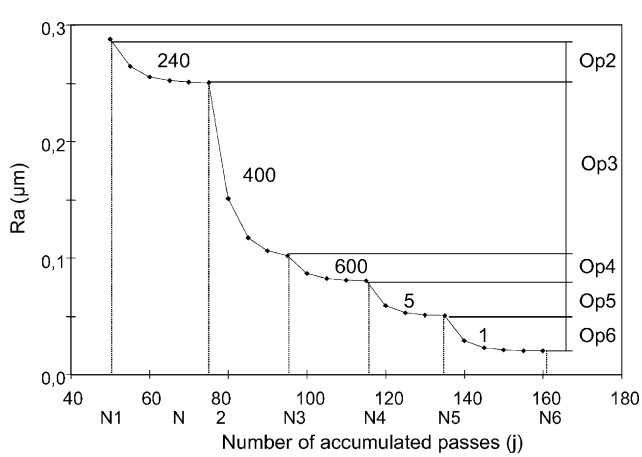


Fig. 10. Theoretical polishing operations (operations 2–6).

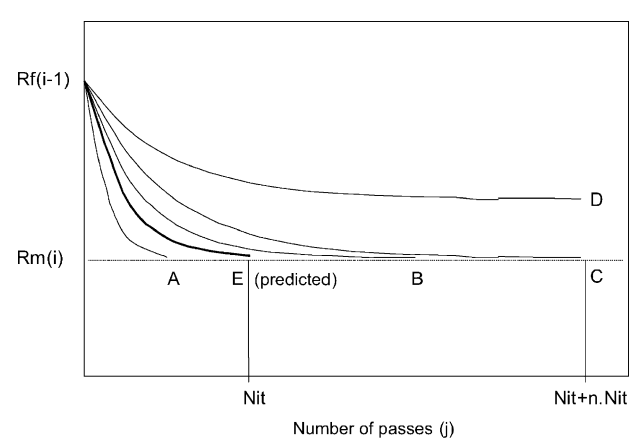


Fig. 12. Deviations from theoretical behavior: evolution models.

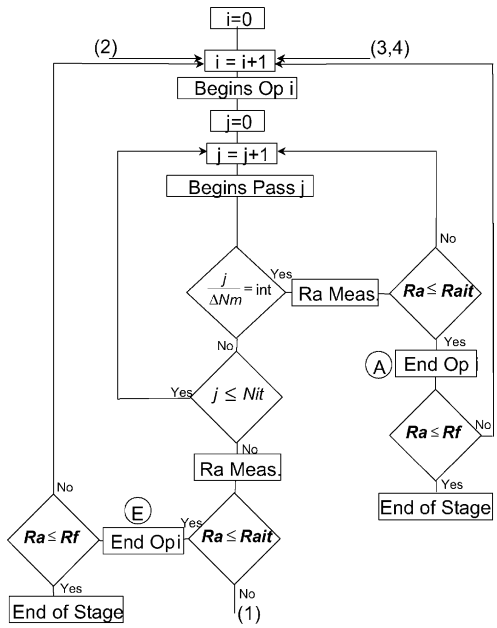


Fig. 13. Inference algorithm for operation control (control of A or E evolution models).

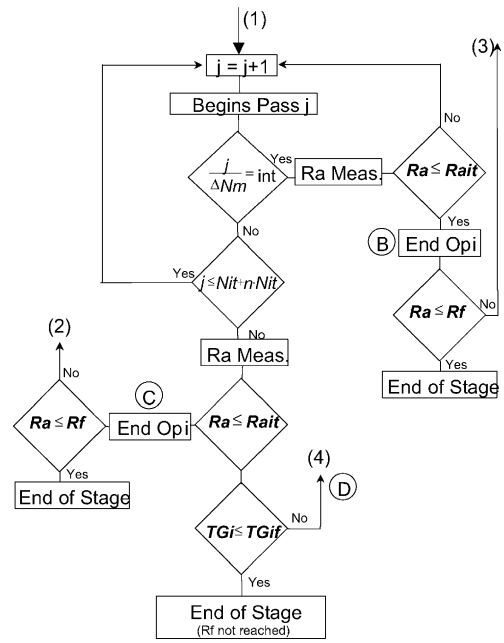


Fig. 14. Inference algorithm for operation control (control of B, C or D evolution models).

The operation control allows the completion of a polishing stage planned by the stage planner. The objective is to follow the stage execution in real time to obtain the best possible finishing state.

The finishing state is determined by measuring the final roughness where it is possible, or alternatively, measuring the brightness of the surface. The former is more precise but can be done only on relatively ample flat surfaces, and it is a method useful for laboratory and research purposes, the latter is more straightforward and it is recommended for industrial applications, mainly due to its superior reliability in these environments.

For optical surface finishing measurements, it can be proven that the index that is more appropriate for metallic surfaces is the geometry of 60°, for which there is a more linear and sensitive response than for the indexes of 85° and 20° in the considered range of surface finishing (Fig. 17).

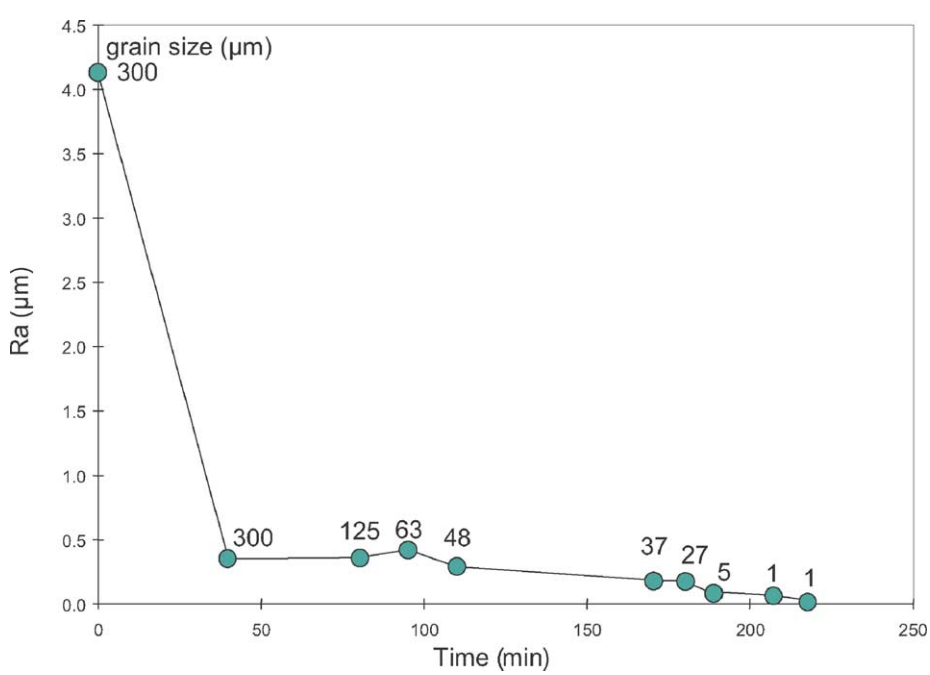


Fig. 15. Evolution process as predicted.

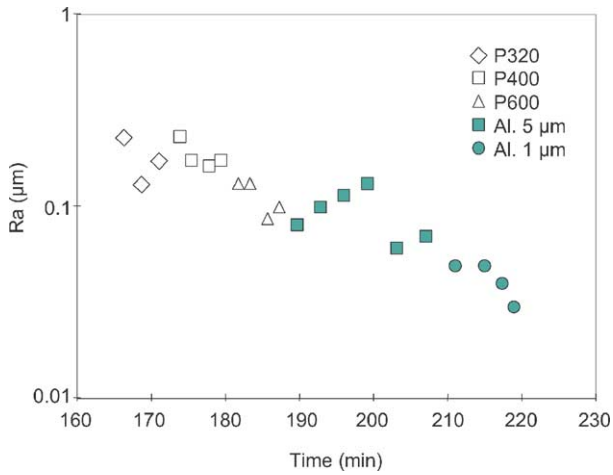


Fig. 16. Evolution process with points out of tune.

For a narrow range of surface finishes and fixed workpiece materials (steel in our case), a relation between the brightness index and roughness for a specified material family, can be derived, and used to correlate laboratory data with industrial applications (Fig. 18).

4.2. Polishing pass control

It is extremely difficult to locate the robot over the surface with the required degree of accuracy. A robotic polishing system based on absolute geometry is clearly not an option due to its impractical nature. On the other hand, a system based in relative geometry requires well-defined dimensions, necessitating the use of a rigid disk, and a model to solve the disk wear problem. Furthermore, a sensor capable of accurately maintaining the distance between the disk and the surface would be needed as well. Such sensors do not have good results in the scenario of tracking the complex three-dimensional trajectories that are found in mold geometry. Even if a perfect sensor existed, the problem of precisely

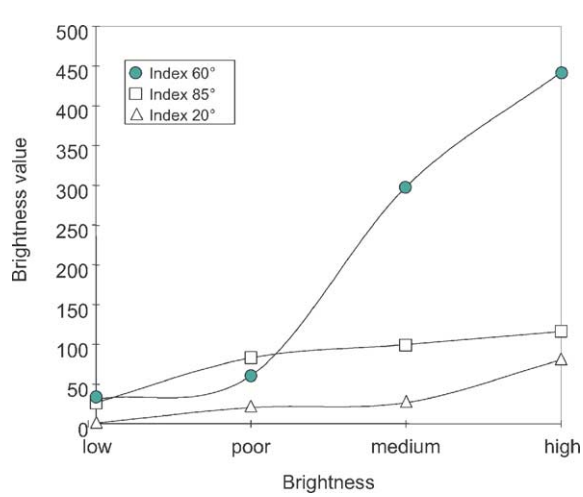


Fig. 17. Evolution of the sensitivity of different brightness indices.

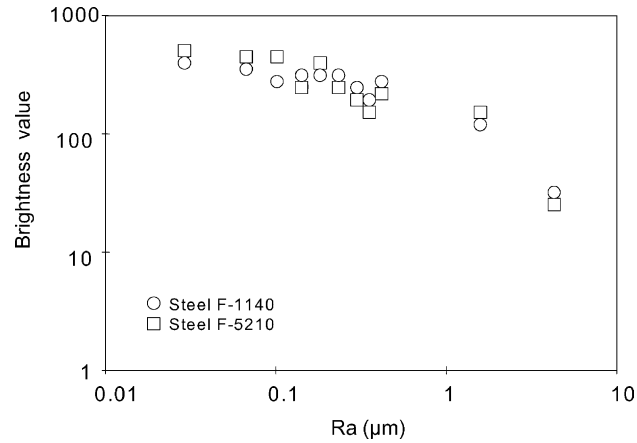


Fig. 18. Experimental relation between roughness and brightness.

locating the contact point and determining its geometry under the polishing wheel is technologically and economically unfeasible with the current technology. Several estimation schemes have been used to estimate the shape and location of the contact area; however the delay and inaccuracies of these estimators have an unacceptable adverse effect on the control response.

Preliminary studies conducted on the polishing process have demonstrated that is possible to carry out the polishing pass control by mean of a contact force control. The force can be controlled by mean of a control loop uncoupled from the robot position control loop. This circumstance requires less accuracy than pure position control-based systems. The S3 control system for ABB robots has a number of adaptive functions available that allow it to execute uncoupled feedback control based on a sensor signal that modifies gains for the position control loop of the robot arm. Fig. 19 presents a scheme to implement force control during the polishing pass.

The reference inputs are the position and force values for the polishing pass. The position is given in a coordinate file that is generated by the CAD automatic programming application. The reference force is determined in the stage

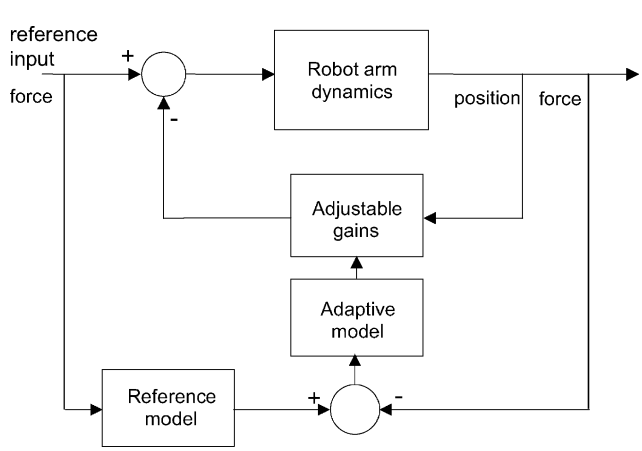


Fig. 19. Force control scheme during the polishing pass.

planner. The force measured in the sensor is the feedback signal that is compared to the reference force value. Finally, the position control loop gains are modified accordingly. Consequently, the robot arm adapts its position accordingly to the force value fixed in the stage planner.

5. Experimental results

Tests have been performed to adjust the parameters of the polishing cell, and to machine parts of different geometries representatives of mold designs for plastic injection.

The cell is composed by an ASEA IRB 3000-6 axis robot, with a load capacity up to 30 kg, equipped with an electrical driven tool with rotational speeds from 0 to 3000 rpm, which can be interchanged with a pneumatic driven tool for high speed with an operational range of 0–18 000 rpm.

The positioning of the tool is done by the code generated from the geometrical CAD information obtained from the part model for which MATRA EUCLID has been used; and the tracking model for the surface to be polished established in the process programming. The movement is made to a feed rate in the range of 100–500 mm/min.

The contact between part and piece is done by the force feedback loop of the robot control, with a value adjusted in the 1–2 N range.

The programming and control of the robot operations is done by a PC486 executing the application written for the conversions from CAD data to robot code, and for the development of the operation and pass controllers.

Several parts with different geometries are polished. The basic shapes are a continuous free-form concave and convex surfaces, and a box type with plane, cylindrical and spherical surfaces.

The parts materials are steel F-1140 (DIN CK45, ASTM 1045) and F-5210 (DIN X210Cr12, ASTM D3) with hardness ranging from 110 to 170 HB, and with typical dimensions ranging from 100 mm × 100 mm to 300 mm × 300 mm, with curvature radius from 10 to 300 mm.

Surface roughness is computed by the R_a value. In the initial state the roughness values are in the 6.0–1.0 μm (N9–N6) range, and in the final state reached values are ranging 0.10–0.05 μm (N3–N2), according to the usual central values obtained in manual polishing operations, 0.20–0.05 μm (N4–N2).

The tool trajectories are obtained from the smooth lines that define the so called ‘tool paths’, combined with a secondary movement called ‘overlapping’. This resembles the traditional motions in manual polishing, done with a linear or curvilinear trajectory over which a secondary orbital motion is done for a more intensive crossover of scratches produced by abrasive grains.

The spacing between tool paths is in the 1–5 mm range, with an overlapping unit dimension equal or greater than this spacing values.

Tools for the initial finishing grades are made of a rubber substrate with abrasive applied on paper, and they are changed to buffing ones with abrasive applied on viscous paste support, in the last finishing grades.

The finished surface in surfaces planes or with a great curvature radius is measured by the profile method with a portable roughness meter Hommel T-500, and with a Dr. Lange Refo 3 brightness meter in the other applicable cases. Measurements in zones with important concavities, acute edges, or difficulties to access, should be measured with a far more sophisticated equipment, with costs not assumable in many of these industrial applications. In consequence, when it is needed, these parts are measured in a metrology laboratory.

The robotic polishing provides several benefits: a more even quality on the surface, avoidance of overpolished zones, not reaching exhaust conditions on the tool life, diminishing the polishing action on summits and increasing it on recessions. By tool change, a more precise adjustment of machining conditions can be achieved. In zones with details of reduced dimension, or high finishing, the robotic polishing can do the rough action, to alleviate the manual machining by reduction of time and effort, allowing a more detailed termination due to the reduction of fatigue levels.

Very good results have been obtained in tracking trajectories over different workpiece geometries automatically programmed from the CAD developed application. The force-contact control system has demonstrated to be a good and economical solution for the problem of dimensional adjusting between virtual CAD geometric positions and real tool positions and orientations in the polishing cell.

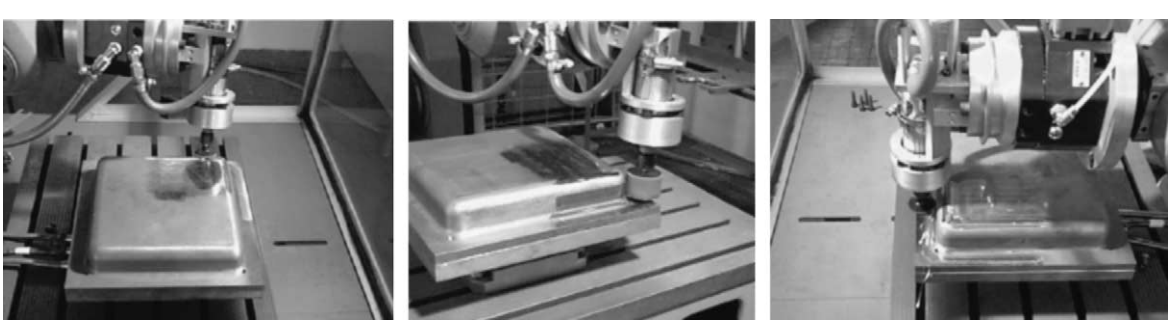


Fig. 20. Different steps in a polishing process.

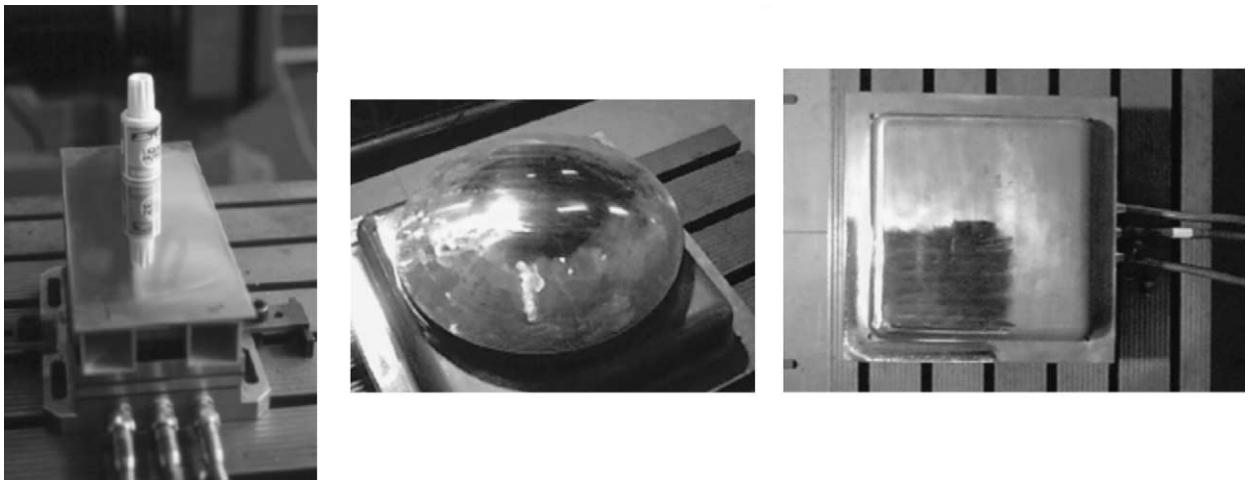


Fig. 21. Polishing operation results.

Fig. 20 depicts different steps in a polishing process for testing tracking of trajectories with force control. The programming system has reduced dramatically the planning and programming time for this kind of operations, improving the productivity and flexibility of the system.

The operation controller has demonstrated a great effectiveness. The main problem for this system still remains in the solution adopted for finishing evaluation. The portable roughness and brightness instruments employed are not designed for automated in-process measuring. Nowadays, development of new instruments designed for these purposes could improve the operation of the controller in order to minimize inspection time.

In Fig. 21 shows the polishing results in three tested workpieces with different geometries: flat, spherical and combined (flat, cylindrical and spherical).

6. Conclusions

In this paper a complete system to polish parts is shown, for application mainly in the mold finishing field with free-form surfaces. Their applicability is for medium or short series where the geometric definition of the part comes from a CAD model. The substitution of traditional manual processes can be done with advantages in time and costs, with usual economical schemes in technological environments and with finishing grades similar to the manual processes. Flexibility in the definition of the tool trajectories over the surface to be polished can modulate the intensity of the abrasive process.

In manufacturing plants with manual polishing operations, the processing time for finishing operations on molds and dies can represent a 17–29% of the total production time [21], the cost percentage in consequence is even higher due to the high hourly cost from the operator who carries out these processes.

The allowable alternatives to reduce this quantity of work that causes such a high cost, can be divided in two groups, those that will improve the state of finishing of the mold at the end of the classic machining process, by means of the employment of machining technologies as the electrodischarge machining (EDM), or the high speed cutting (HSC); and those that automate as far as possible the finishing operations (like polishing) that now are carried out in a manual way.

The alternatives are not exclusive, since when such technologies are used for dies and mold machining, a later better finishing could be necessary, and this can only be obtained by polishing.

It can be that the automated alternative with manual programming throws a reduction in cost of only 10% due to the incidence of high hourly cost of the programmer, however, in the alternative with automatic programming the reduction is of about 25%, due to the reduction of the time needed for the robot programming, which supposes a considerable reduction that justifies this system from the economic point of view.

The automation of a complex process like polishing requires the research of the whole problem if it intends obtain a reliable solution. The main goal of this work is to integrate very different aspects and obtain a global modeling that allows the different elements of the process to function hierarchy without problems of adaptation. For example, theoretical phases, like process and stage planners are fully integrated with very experimental phases like operation and pass controllers.

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